ECLT5930B/SEEM5740B Engineering Economics and Cost Management

Chapter 9
Price Changes

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Inflation and deflation

- General price inflation: an increase in the general price level of goods and services in an economy.
  - To obtain the same product or service, a higher amount of money is paid.
- General price deflation: a decrease in the general price level of goods and services in an economy.
  - To obtain the same product or service, a lower amount of money is paid.

Is inflation or deflation bad?

- High or hyper inflation is undesirable because it reduces the quality of life for the general public.
  - The annual inflation rate in year 2016 for Venezuela hits 800%.
- Most central banks target at low and stable inflation rate.
  - The central bank in the US targets annual inflation rate at 2%.
- Deflation is generally believed to be bad because it is associated with economic recession.

Inflation measures by price index

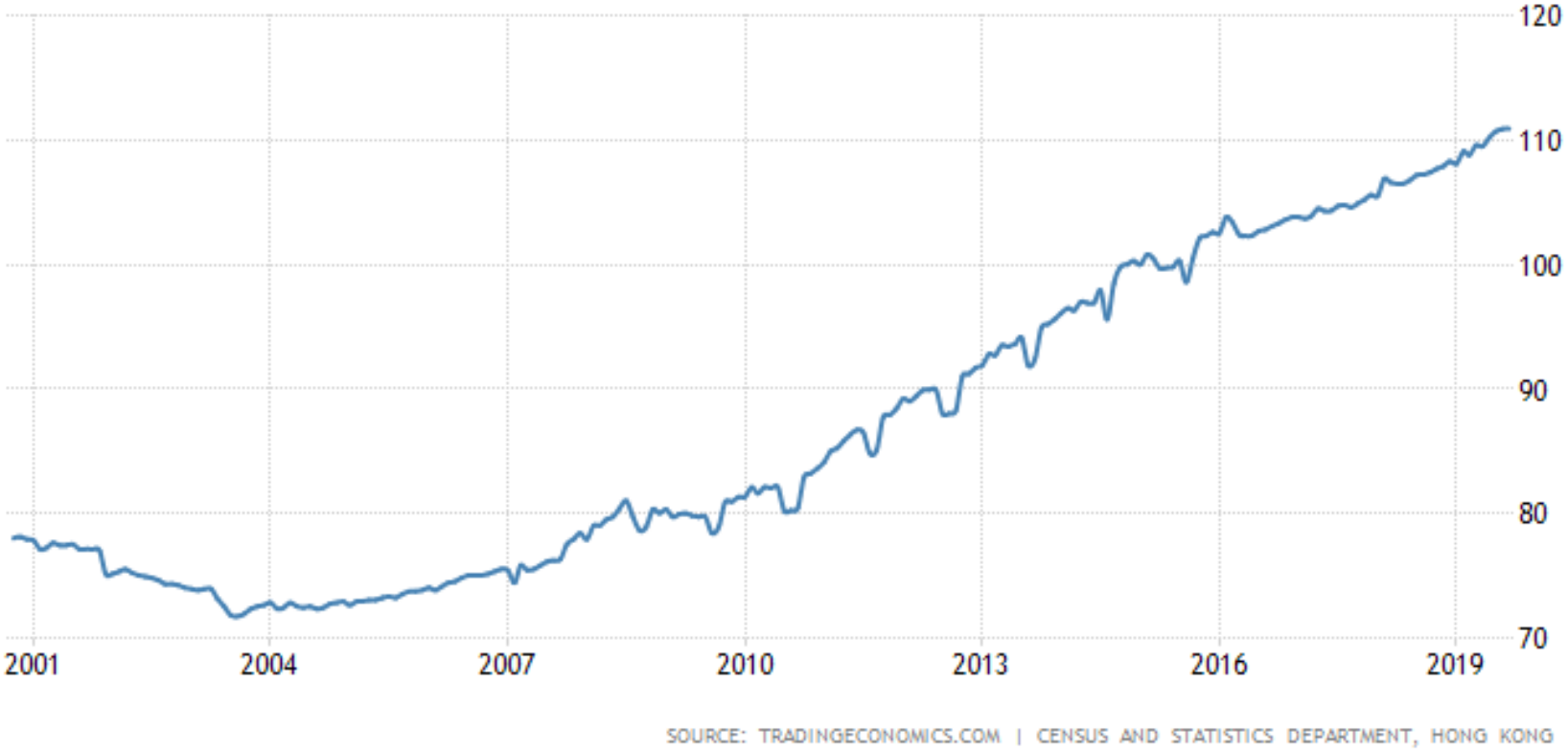
- A price index is a dimensionless numerical value which is used to measure price level for a group of goods and services.
- Popular price indexes:
  - (1) Consumer price index (CPI): for consumers.
  - (2) Producer Price index (PPI): for producers.
- From CPI (PPI) we can obtain the general price inflation rate for goods and services for consumers (producers).

Inflation rate from the price index

$$(CPI\ Annual\ Changing\ Rate)_k = \frac{(Index)_k - (Index)_{k-1}}{(Index)_{k-1}}$$

- The CPI index was 218 and 225 at the end of 2010 and 2011 respectively. What is the general inflation rate for consumers in 2011?
- Solution:
  - (225-218)/ 218 = 3.21%. This means the price increases by 3.21% in year 2011 compared to 2010.
- Remark: Negative value means deflation.

CPI of Hong Kong



The purchase power of money

- Purchasing power of A dollars is the amount of goods and services that can be purchased with A dollars.
- In general, inflation reduces the purchase power of money.
  - Suppose at base time, the purchasing power of $1 is 1 unit of goods and services.
  - Now suppose the inflation rate is 2%. One period after the base time, the price of 1 unit of goods and service increases to 1 + 2% = $1.02. Hence the purchasing power of $1 at b+1 becomes 1/1.02≈ 0.98 units of goods and services.
  - To buy the same amount of goods and services at b+1, you need $1.02.

Terminology

- Actual dollars (A$)
  - The number of dollars associated with a cash flow as of the time it occurs
- Real dollars (R$)
  - Values expressed in terms of the same purchasing power relative to a particular time
- Mike finds a job. By contract, his annual salary is $300,000 for the following 3 years
  - In A$: it is $300,000 for each of the following year
  - In R$: it varies from year to year, depending on CPI

Inflation rate

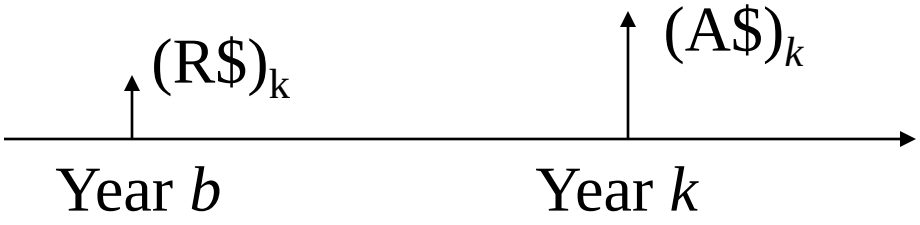
- General price inflation rate, denoted by *f*.
  - The CPI annual changing rate
  - For simplicity, we may assume *f* is the same for every year in a period of time, referred to as the (average) inflation rate
  - Inflation rate for a specific year can also be defined

Year	1	2	3	4
CPI	504	538.4	577	629.5
Inflation rate (%)		6.83	7.17	9.10

$$f = (629.5/504)^{1/3} - 1 = 7.69\%$$

A$, R$ and inflation rate

- Given $(A\$)_k$ at year k, inflation rate f, a base year b
  - The equivalent $(R\$)_k = (A\$)_k\,(P/F, f\,\%, k-b)$
  - Note b=0 is the current time, b=1 is the end of current year (year 1)
  - $R = \frac{A}{(1+f)^{k-b}}$



Example 9.1

- Suppose my salary is $50,000 in the current year
- It will increase 5% per year through year 4
  - In terms of actual dollars
- The general inflation rate is 8%
- What is the salary in terms of the real dollars of year 1?

Year	1	2	3	4
A$	50000	52500	55125	57881
R$($b$=1)	50000	48611	47261	45948

52500(P/F,8%,1) 55125(P/F,8%,2)

Market vs. real interest rate

- Market interest rate (i_m):
  - gives how many actual dollars can be earned in a period if $1 is invested now (also called "nominal interest rate").
- Real interest rate (i_r):
  - gives how many real dollars can be earned in a period if $1 is invested now
- Remark: The interest rates you observed in life are almost always market interest rate.

Market vs. real interest rate

- Relationship
  - If we invest A_b ($A_b = R_b$) now, what will be its worth in actual dollars at the end of period b + 1?
  - $A_b(1 + i_m) = A_b(1 + i_r)(1 + f)$
  - $i_m = i_r + i_r f + f \quad or \quad i_r = \frac{i_m - f}{1+f}$
- When f is small, we have the following approximation that works well
  - $i_r \approx i_m - f$

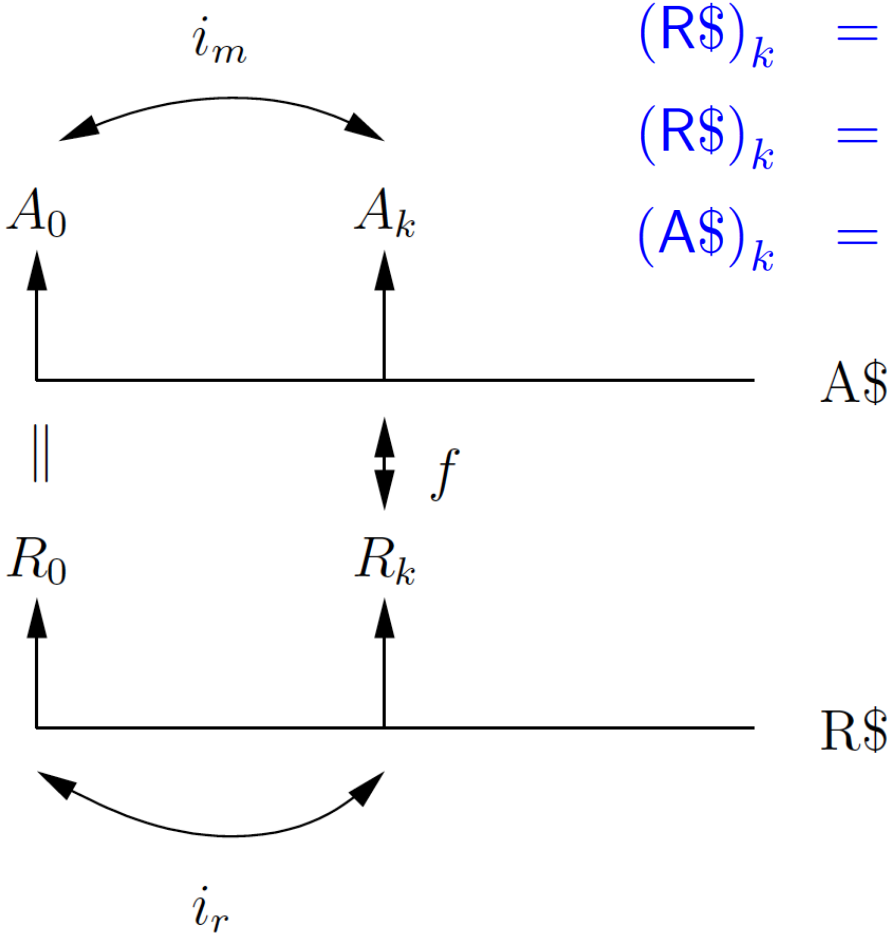
Depositing money in the bank

- One-year time deposit rate in Hong Kong is absolutely less than 1%.
- The current forecast for inflation rate in Hong Kong is around 2.5%.
- The real interest rate offers by the bank is negative. If you deposit your money in the bank, your money is losing purchasing power over time.
- In practice, you need to find investments with return that beats the inflation rate.

Which interest rate to use

- We can calculate PW with either A or R, but we must use the appropriate interest rate.
  - If the cash flows are in actual dollars, then use i_m,
  - If the cash flows are in real dollars, then use i_r.
- These two approaches give the same result.
- The calculated PW is denominated in terms of real dollars (assuming time 0 is the base time).
- IRR (ERR)
  - Nominal IRR (ERR): the actual-dollar cash flows;
  - Real IRR (ERR): the real-dollar cash flows.

Which interest rate to use



$$
\begin{aligned}
(\mathsf{R\$})_k &= (\mathsf{A\$})_k \times (P/F, f\%, k-b), \\
(\mathsf{R\$})_k &= (\mathsf{R\$})_b \times (F/P, i_r\%, k-b), \\
(\mathsf{A\$})_k &= (\mathsf{R\$})_b \times (F/P, i_m\%, k-b),
\end{aligned}
$$

Example 9.1 revisited

- What is the PW of the (a) actual-dollar and (b) real-dollar cash flows at the end of year 1?
  - $MARR_m$ = 10% per year, b=1, and f=8%.

Year	1	2	3	4
A$	50000	52500	55125	57881
R$($b$=1)	50000	48611	47261	45948

Example 9.1 revisited

- Solution
  - $PW_A(10\%) = 50000 + 52500(P/F, 10\%, 1) + 55125(P/F, 10\%, 2) + 57881(P/F, 10\%, 3) = 186772$
  - $MARR_r = \frac{MARR_m - f}{1+f} = \frac{0.1-0.08}{1.08} = 1.852\%$
  - $PW_R(1.852\%) = 50000 + 48611(P/F, 1.852\%, 1) + 47261(P/F, 1.852\%, 2) + 45948(P/F, 1.852\%, 3) = 186772$

Year	1	2	3	4
A$	50000	52500	55125	57881
R$($b$=1)	50000	48611	47261	45948

Example 9.2

- You would like to have $1M dollars in today's purchasing power in your retirement savings account 20 years from now. If the general inflation rate is expected to be 5% per year and the savings account earns 10% interest per year,
  - what lump sum amount should you deposit now?
  - what equal end-of-year amounts should you deposit into the savings account for 20 years?

Example 9.2

- Solution
  - The 10% earned by the savings account is market interest rate. So i_m = 10%.
  - \$1M is equivalent to $1{,}000{,}000(1.05)^{20}$ = 2,653,298 in actual dollars. The lump sum is equal to
    - $2{,}653{,}298 \times (P/F, 10\%, 20) = 394{,}395.8$
  - The equal end-of-year amounts are actual dollars.
    - $A = 2{,}653{,}298 \times (A/F, 10\%, 20) = 46{,}325.6$

Example 9.3

- A machine costs $180,000 with useful life of 10 years
  - Annual revenue in R$: $36,000; Salvage value in R$: $30,000; Average inflation rate: 8%; Effective tax rate: 38%
  - Annual maintenance cost for the first 3 years (A$): $2,800
  - Depreciation is for 5 years, with (A$)
    - 36,000; 57,600; 34,560; 20,736; 20,736; 10,368
- Question
  - Based on an actual-dollar after-tax analysis, is this capital investment justified?
  - Develop the ATCF in real dollars.

Example 9.3

- Analysis under MARR=15%

End of Year (k)	(1) A$ Cash Flows	(2) Contract (A$)	(3) BTCF (A$)	(4) Depreciation (A$)	(5) Taxable Income	(6) Income Taxes ($t = 0.38$)	(7) ATCF (A$)	(8) R$ Adjustment $[1/(1+f)]^{k-b}$	(9) ATCF (R$)
0	−$180,000		−$180,000				−$180,000	1.0000	−$180,000
1	38,880[a]	−$2,800	36,080	$36,000	$80	−$30	36,050	0.9259	33,379
2	41,990	−2,800	39,190	57,600	−18,410	+6,996	46,186	0.8573	39,595
3	45,349	−2,800	42,549	34,560	7,989	−3,036	39,513	0.7938	31,366
4	48,978		48,978	20,736	28,242	−10,732	38,246	0.7350	28,111
5	52,895		52,895	20,736	32,159	−12,220	40,675	0.6806	27,683
6	57,128		57,128	10,368	46,760	−17,769	39,359	0.6302	24,804
7	61,697		61,697		61,697	−23,445	38,252	0.5835	22,320
8	66,632		66,632		66,632	−25,320	41,312	0.5403	22,320
9	71,964		71,964		71,964	−27,346	44,618	0.5003	22,320
10	77,720		77,720		77,720	−29,534	48,186	0.4632	22,320
10	64,767[b]		64,767		64,767	−24,611	40,156	0.4632	18,600

[a] $(A\$)_k = \$36{,}000(1.08)^{k-0}, k = 1, \ldots, 10.$
[b] $MV_{10,A\$} = \$30{,}000(1.08)^{10} = \$64{,}767.$

36000(F/P,8%,n)

A$ already

PW for A$ (15%)= 33,790

PW analysis from R$ (6.481%)= 33,790

$$MARR_r = \frac{MARR_m - f}{1+f} = \frac{0.15-0.08}{1+0.08} = 0.06481$$

Exchange rate with foreign currency

- The exchange rates with foreign currencies change from time to time
  - Using HK dollars as the base
  - The exchange rate is represented by the amount of foreign currency needed to buy 1 HK dollar
- f_e: exchange rate change for a foreign currency to HK dollars over time
  - Analogous to the inflation rate (HK$ as R$)
  - $f_e > 0$: the foreign currency is de-valued
  - $f_e < 0$: the Hong Kong dollar is de-valued

Exchange rate with foreign currency

- The money of country A
  - Present exchange rate: 10.7 units per HK dollar
- If the money of country A is de-valued with an exchange rate change of 4.2% per year subject to HK$
  - f_e = 4.2%
- After 3 years, the exchange rate becomes

$$10.7*(1.042)^3=12.106$$

Cash analysis with foreign money

- Cash flow with respect to HK dollars
  - Interest rate with HK dollars: i_{HK}
- Cash flow with respect to foreign currency
  - Interest rate with foreign money: i_{FC}
- Given the exchange rate change f_e for the foreign money

$$(1 + i_{HK})(1 + f_e) = (1 + i_{FC})$$

- So,

$$i_{FC} = i_{HK} + f_e + f_e i_{HK}$$

$$i_{HK} = \frac{i_{FC} - f_e}{1 + f_e}$$

Example 9.4

- A HK company is considering a potential project in another country A.
  - Current exchange rate is 425 units of A = 1 HK$
  - It is expected that currency A is to be devalued of 2% per year subject to HK$
- In currency A, the project
  - Initial investment is 850M
  - Annual profit is 270M for 6 years
  - Salvage value is 120M
- The MARR in HK$ is 20%

Example 9.4

- Solution
  - MARR in HK$ is 20%
  - MARR in foreign currency is

    $i_{HK}+f_e+f_e i_{HK}=20\%+2\%+(20\%)(2\%)=22.4\%$
  - PW for the cash flow in foreign currency

    $-850+270(P/A,22.4\%,6)+120(P/F,22.4\%,6)=32.59M$
  - PW in terms of HK$ is 32.59/425=0.0767M

Example 9.4

- IRR in foreign currency
  - By Excel function, we have IRR(currency A)=23.92%
  - IRR(HK$)= $(i_{FC}-f_e)/(1+f_e)$=(23.92% −2%)/102%
    =21.5%
- Cash flow analysis in HK dollars

Foreign currency	-850	270	270	270	270	270	390
exchange rate	425	433.5	442.17	451.0134	460.0337	469.2343	478.619
HK dollar	-2	0.622837	0.610625	0.598652	0.586914	0.575405	0.814844

IRR for cash flow in HK$ is 21.5%

PW for cash flow in HK$ and MARR=20% is 0.0767

- Chapter 8 of the textbook